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Paper Code : PH-301

PHYSICS-II

Time Allotted : 3 Hours | *Full Marks* : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP – A
(Multiple Choice Type Questions)

1. Choose the correct alternatives for any *ten* of the following : $10 \times 1 = 10$

   i) The degree of freedom of particle constrained to move on a surface of sphere is

   a) 1 b) 2

   c) 3 d) 4.

   ii) If $\vec{\alpha} = 33\hat{i} - 2\hat{j} + \hat{k}$ and $\vec{\beta} = 2\hat{i} - \hat{k}$, then $(\vec{\alpha} \times \vec{\beta}) \cdot \vec{\alpha}$ is equal to

   a) 0 b) 2

   c) 29 d) $\hat{i} + 3\hat{k}$.

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iii) The Lagrangian of a system is a function of

a) $\dot{q}_i,\ \dot{p}_j,\ t$ b) $q_j,\ \dot{p}_j,\ t$

c) $\dot{q}_j,\ p_j,\ t$ d) $q_j,\ \dot{q}_j,\ t.$

iv) Average energy $< E >$ of an electron in a metal at $T = 0$ K is

a) E_F b) $E_F/2$

c) $\frac{3}{5}E_F$ d) $\frac{3}{2}E_F.$

v) The continuity equation is given by

a) $\nabla.J + \frac{\partial \rho}{\partial t} = 0$ b) $\nabla.J = 0$

c) $\frac{\partial \rho}{\partial t} = 0$ d) $\nabla \times J = 0.$

vi) For $T > 0$ K, the probability of occupancy of an electron at the Fermi energy level is

a) 1/2 b) 0

c) 1 d) 1/4.

vii) If $\psi\ (x,\ t)$ is a normalized one dimensional wave function, then

a) $\int_{-\infty}^{+\infty} \psi^* \psi\, dx = \frac{1}{\sqrt{2}}$

b) $\int_{-\infty}^{+\infty} \psi^* \psi\, dx = 1/2$

c) $\int_{-\infty}^{+\infty} \psi^* \psi\, dx = 0$

d) $\int_{-\infty}^{+\infty} \psi^* \psi\, dx = 1$

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viii) At any position the vector potential is given by $\vec{A} = 5(x^2 + y^2 + z^2)\,\hat{i}$, the magnetic field at that position will be

a) $x\hat{i} + y\hat{j}$ b) $5(x\hat{j} + y\hat{k})$

c) $10(z\hat{j} + y\hat{k})$ d) 0.

ix) The Pointing vector is given by

a) $\vec{H} \times \vec{E}^*$ b) $\vec{H} \cdot \vec{E}$

c) $\frac{1}{2}\vec{E} \times \vec{H}^*$ d) $\frac{1}{2}\left(\vec{E} \times \vec{H}^*\right)$.

x) The number of quantum states $g(E)\,dE$ depends on the energy as

a) $E^{1/2}$ b) $E^{3/2}$

c) E d) E^2.

xi) Which of the following pairs of observables cannot be measured simultaneously ?

a) $\hat{p}_x \cdot \hat{p}_y$ b) $\hat{x} \cdot \hat{p}_y$

c) $\hat{y} \cdot \hat{p}_y$ d) $\hat{x} \cdot \hat{y}$.

xii) The velocity of plane electromagnetic wave is given by

a) $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$ b) $c = \frac{1}{\mu_0 \epsilon_0}$

c) $c = \frac{\mu_0}{\epsilon_0}$ d) $c = \frac{\epsilon_0}{\mu_0}$.

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xiii) Operator corresponding to the kinetic energy is

a) $\frac{3\hbar^2}{2m}\nabla^2$ b) $-\frac{\hbar^2}{2m}\nabla^2$

c) $\frac{2\hbar}{m}\nabla^2$ d) $-\frac{2m}{\hbar^2}\nabla^2$.

xiv) MB statistics is applied for

a) Ideal gas b) Photons

c) Electrons d) None of these.

xv) The unit of polarizability is

a) Nm^2/c^2 b) debye

c) Fm^2 d) V/m.

GROUP – B

(Short Answer Type Questions)

Answer any *three* of the following. $3 \times 5 = 15$

2. a) Prove that $\vec{\nabla}\phi \cdot d\vec{r} = 0$, symbols have their usual significance. 2

b) Find the normal unit vector at the point $\left(\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}}, \frac{c}{\sqrt{3}}\right)$ on the surface of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$. 3

3. a) What do you mean by cyclic coordinate ? Explain it. 1 + 2

b) Find the magnetic field at the centre of a semicircular conducting loop of radius 3 cm carrying current $\frac{1}{\pi}$ amp. 2

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4. Write down the Schrödinger equation for one dimensional motion of a free particle in one dimensional potential box. Find its eigenfunction and eigenenergy.

5. Find the Lagrnge's equation of motion of freely falling particle from a height H. Show that the plane of reference does not have any effect on equation of motion. 3 + 2

6. a) Two similar balls of mass m are hung from silk threads of length l and carry same charges. Prove that for a small angle θ. the separation of the charges. will be

$$x = \left(\frac{q^2 l}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$$

b) If a charged article of charge 0·5 C is moving with a velocity $\vec{v} = 4\hat{i} - \hat{j} + 2\hat{k}$ m/s through an electric field $\vec{E} = 10\hat{i} + 5\hat{k}$ and magnetic flux density $\vec{B} = 2\hat{i} + 3\hat{j} - 5\hat{k}$. then find Lorentz force. 3 + 2

GROUP – C

(Long Answer Type Questions)

Answer any *three* of the following. 3 × 15 = 45

7. a) If $\vec{v} = \vec{\omega} \times \vec{r}$, prove that $\vec{\omega} = \frac{1}{2} curl\ v$.

b) Using Gauss' law find out the electric field due to a charge q uniformly distributed over a sphere of radius R (i) at an external point, (ii) at an internal point and (iii) at the surface of the sphere. Show the variation graphically.

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c) What is dielectric ? What is dielectric constant ? Show that $\vec{D} = \varepsilon_0 \vec{E} + \vec{P}$. 3 + (5+1) + (1 + 2 + 3)

8. a) Find magnetic field at a point (1, 1, 1) if vector potential at that position is $\vec{A} = (10x^2 + y^2 - z^2)\,\hat{j}$.

b) A very long cylinder of radius a carries current I. which is uniformly distributed over the cross-section of the cylinder. If the current flows along the axis of the cylinder then find out the electric field in the conductor, the magnetic field just outside the conductor and the Poynting vector at the surface of the cylinder. You may assume that the conductivity of the material is σ.

c) Calculate the magnetic field at the centre of a circular lamp carrying current I.

d) If $\vec{E} = \vec{E}_{e^{i(kx-\omega t)}}$ denotes the electric vector of an electromagnetic field in vacuum then find out the magnetic vector. 2 + (2 + 2 + 2) + 5 + 2

9. a) Define Ampere's law in integral form and its conversion to differential form.

b) If $\vec{A} = \frac{1}{2}(\vec{B} \times \vec{r})$. show that $\vec{A}$ is the magnetic vector potential associated with $\vec{B}$.

c) A solenoid has 4 layers of 1200 turns each. Its length and mean radius are 3m and 0·25 m respectively. Find the magnetic field and flux at its centre. if a current of 2·5 A flows through it.

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d) Write down Maxwell's equations in electromagnetic theory and interpret them. (1 + 2) + 3 + 3 + (2 + 4)

10. a) The Lagrangian of a system is given by $L = \frac{1}{2}\alpha \dot{q}^2 + \frac{1}{2}\beta q$, where α and β are constants. Obtain Lagrange's equation of motion and also find the Hamiltonian of the system.

b) What is meant by cyclic coordinate ? Show that if a given coordinate is cyclic then the generalize momentum corresponding to the cyclic coordinate is constant of time.

c) Prove that for a conservative system, the Hamiltonian represents the total energy of the system.

d) Obtain the Hamiltonian of a simple harmonic oscillator moving in one dimension.

(2 + 3) + (2 + 2) + 3 + 3

11. a) What are the basic postulates of quantum mechanics ? Starting from the time independent Schrödinger equation for 1-D motion of a particle in 1-D potential box, show that the energy eigenvalues and momentum eigenvalues of the particle are quantized in the 1-D potential box. Also find out the normalized wave function for that particle. Plot φ and $|\varphi|^2$ as a function of position for 1-D box.

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b) Wave function of a quantum mechanical system is $\varphi = \frac{1}{\sqrt{3}}\varphi_1 + \frac{\sqrt{2}}{\sqrt{3}}\varphi_2$. What is the probability of the system being in φ_2 state? (3 + 6 + 2 + 2) + 2

12. a) Give a comparative study of *M-B*, *B-E* and *F-D* statistics.

b) Using *F-D* statistics find total number of free electrons in a metal at absolute zero (0 K) temperature. Hence find out the mean internal energy of free electrons of metal at 0 K.

c) If the Fermi energy level of a metal is 12 eV at absolute zero temperature, find out the average energy of free electrons. 5 + (5 + 3) + 2

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